PTM MODULE (PARENT-TEACHER MEETING ENGINE)

SmartCampus School Management System
Technical Specification & Implementation Guide — v1.0 • March 2026
Confidential • Internal Documentation

Audience: School Administrators • Teachers • IT Team • Parents
Related Modules: Student Management • Communication • Calendar • Notifications • Parent Portal

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1. MODULE OVERVIEW

1.1 Purpose

The PTM Module facilitates structured scheduling, management, and tracking of Parent-Teacher Meetings. It enables schools to:

- Organize PTM sessions by term or event
- Allow parents to book meeting slots with teachers
- Manage teacher availability and schedules
- Send automated reminders and notifications
- Track attendance and feedback
- Provide analytics on engagement and meeting outcomes

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2. CORE OBJECTIVES

1. Enable structured scheduling of parent-teacher meetings
2. Provide seamless booking experience for parents
3. Manage teacher availability and time slots efficiently
4. Automate notifications and reminders
5. Track attendance, feedback, and engagement
6. Ensure secure, role-based access to meeting data
7. Support mobile-first and accessible design
8. Integrate with calendar and communication systems

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3. REAL-WORLD SYSTEM FLOW

3.1 Lifecycle Flow

PTM Session Created → Teacher Slots Defined → Parent Booking Opened → Slots Reserved → Notifications Sent → Meetings Conducted → Attendance Logged → Feedback Collected → Reports Generated

3.2 Daily Operational Flow

Admin creates PTM session
↓
Teachers define available time slots
↓
Parents log in and book preferred slots
↓
System confirms booking and sends notifications
↓
Meeting conducted (physical or virtual)
↓
Attendance and notes recorded
↓
Feedback collected and analytics updated

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4. MODULE ARCHITECTURE

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PTM Module
│
├── PTM Session Management
├── Slot & Availability Management
├── Booking System
├── Notifications & Reminders
├── Meeting Tracking & Attendance
├── Feedback & Surveys
├── Analytics & Reporting
├── Calendar Integration
├── Parent & Teacher Portals
├── Mobile & Accessibility Support
├── Security & Role-Based Access
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5. PTM SESSION MANAGEMENT

5.1 Features

- Create PTM sessions (term-based or special events)
- Define session dates, duration, and format (physical/virtual)
- Activate/deactivate booking windows
- Track session status (upcoming, ongoing, completed)

5.2 Data Model

PTMSession

- session_id
- title
- term
- start_date
- end_date
- booking_deadline
- status

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6. SLOT & AVAILABILITY MANAGEMENT

6.1 Features

- Teachers define available meeting slots
- Customizable duration per slot (e.g., 10–30 minutes)
- Prevent double booking
- Bulk slot creation

6.2 Data Model

Slot

- slot_id
- teacher_id
- session_id
- start_time
- end_time
- status (available/booked/cancelled)

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7. BOOKING SYSTEM

7.1 Features

- Parents book slots based on availability
- Multi-child booking support
- Real-time availability updates
- Booking confirmation and cancellation

7.2 Data Model

Booking

- booking_id
- parent_id
- student_id
- teacher_id
- slot_id
- status (confirmed/cancelled/completed)

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8. NOTIFICATIONS & REMINDERS

8.1 Features

- Automated email/SMS/push notifications
- Booking confirmations and reminders
- Alerts for schedule changes or cancellations

8.2 Triggers

- Slot booked
- Upcoming meeting (24h/1h reminders)
- Slot cancelled or rescheduled

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9. MEETING TRACKING & ATTENDANCE

9.1 Features

- Mark attendance for parents and teachers
- Record meeting notes and outcomes
- Track no-shows and reschedules

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10. FEEDBACK & SURVEYS

10.1 Features

- Post-meeting feedback forms
- Parent and teacher satisfaction ratings
- Suggestions and comments

10.2 Data Model

Feedback

- feedback_id
- booking_id
- rating
- comments
- submitted_by

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11. ANALYTICS & REPORTING

11.1 Features

- Number of meetings scheduled vs completed
- Attendance rates and no-show analysis
- Teacher engagement metrics
- Parent participation trends

11.2 Dashboards

- Session summary overview
- Teacher performance and availability
- Parent engagement insights

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12. CALENDAR INTEGRATION

12.1 Features

- Sync meetings with Google Calendar / Outlook
- Add-to-calendar functionality for parents and teachers
- Real-time updates for schedule changes

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13. PARENT & TEACHER PORTALS

13.1 Features

- Parents: Book, view, and manage meetings
- Teachers: Manage availability, view bookings, add notes
- Admin: Full control and reporting

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14. MOBILE-FIRST & ACCESSIBILITY DESIGN

14.1 Requirements

- Fully responsive booking interface
- Mobile-friendly notifications and reminders
- Accessibility compliance (screen readers, contrast, navigation)

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15. SECURITY & ROLE-BASED ACCESS

Role	Access Scope
Admin	Full session management and reporting
Teacher	Manage slots, view bookings, add notes
Parent	Book and manage own meetings
Student	Optional view-only access

15.2 Security Features

- Secure authentication and session management
- Encrypted data transmission
- Audit logs for all bookings and changes

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16. SCALABILITY

- Support large schools with hundreds of teachers and parents
- Efficient slot management and booking handling
- Multi-campus support

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17. IMPLEMENTATION GAPS (CRITICAL FIXES)

17.1 Current Issues Identified

- Dashboard stuck on loading state (poor UX feedback)
- No analytics or visual insights
- Missing communication and notification system
- Lack of audit logs and reporting
- Limited customization of slots and sessions

17.2 Required Fixes

- Improve UI feedback (loading states, tooltips)
- Implement analytics dashboards
- Add notification engine (SMS/email/push)
- Introduce audit trails and logs
- Enable flexible slot customization

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18. FUTURE ENHANCEMENTS

- Video conferencing integration (Zoom/Google Meet)
- AI-based scheduling optimization
- Smart reminders based on user behavior
- Integration with payment systems (premium consultations)
- Advanced reporting and predictive engagement analytics